Solving the follwing system: E-Vex

$$\begin{cases} \ln x + \ln y = 0 \\ x + y = \frac{10}{3} \end{cases}$$

$\ln x + \ln y \Rightarrow \ln(x \cdot y) = 0$

$e^{\ln(x \cdot y)} = e^0$ then $\begin{cases} x \cdot y = 1 \\ x + y = \frac{10}{3} \end{cases}$

$x \cdot y = 1$

$$\boxed{y = \frac{10}{3} - x}$$

$$x\left(\frac{10}{3} - x\right) = 1$$

$$\frac{x10}{3} - x^2 - 1 = 0 \xRightarrow{*-1} x^2 - \frac{x10}{3} + 1 = 0$$

then $\Rightarrow 3\left(x^2 - \frac{x10}{3} + 1\right) = 3(0)$

$$3x^2 - 10x + 3 = 0$$

$x = 3$, $x = \frac{1}{3}$ $\Rightarrow 0 < \frac{1}{3} < 3$ acc

when $x = 3$: $3y = 1$
$y = 3$

when $x = \frac{1}{3}$
$\frac{1}{3}y = 1 \Rightarrow y = \frac{1}{3}$

$$S: \left(3, \frac{1}{3}\right) \left(\frac{1}{3}, 3\right)$$

E-Vex